Bihar Engineering University, Patna

B. Tech 6th Semester Examination, 2024

Course : B.Tech
Code : 101603

Subject: Design of Steel Structures

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer the following (Any seven question only): **[2 x 7 = 14]**

(a) High strength bolts are designed based on
(i) Friction (ii) Tension (iii) Compression (iv) Shear

(b) A steel plate is 30cm wide and 10mm thick. If the diameter of the bolt hole is 20mm, the net section area of the plate is
(i) 18cm2 (ii) 280cm2 (iii) 28cm2 (iv) 32.42cm2

(c) The slenderness ratio in a tension member as per IS code where reversal of stress is due to loads other than wind or seismic should not exceed
(i) 350 (ii) 180 (iii) 100 (iv) 60

(d) Which of the following is not a compression member?
(i) Strut (ii) Tie (iii) Rafter (iv) Boom

(e) The effective length of a battened column can be increased by
(i) 2 % (ii) 10 % (iii) 5 % (iv) 15 %

(f) For the block shear failure of a tension member, the failure occurs along a path through the connection involving
(i) Tension on the two perpendicular planes
(ii) Shear on the two perpendicular plane
(iii) Tension and shear on the same plane
(iv) Tension on one plane and shear on another plane

(g) Critical bending moment capacity of a beam undergoing lateral torsional buckling is a function of
(i) Pure and warping torsional resistance
(ii) Warping torsional resistance only
(iii) Pure torsional resistance only
(iv) Does not depend on anything

(h) The load moment interaction diagrams are more or less similar to
(i) Reinforced concrete column
(ii) Aluminum Columns
(iii) Prestressed Columns
(iv) Steel columns

(i) The pitch of spirals is limited to a range of
(i) 25 – 75 mm (ii)40–45 mm (iii) 15 – 30 mm (iv) 10 – 30 mm

(j) Which of the following is not correct?
(i) Angles and T section are strong in bending
(ii) Channels can be used only for light loads
(iii) I sections are most efficient and economical shapes
(iv) I section with cover plates are provided when large section modulus is required

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Q.2 (a) Explain the key material properties of steel used in structural design and how they influence the behavior of steel structures. [7]

(b) Discuss how partial safety factors and load combinations are applied in the analysis and design of steel structures according to relevant codes and standards. [7]

Q.3 (a) Design a lap joint between two plates as shown in figure 1 below to transmit a factored load of 70 kN using M 16 bolts of grade 4.6 and grade 410 plates. [7]

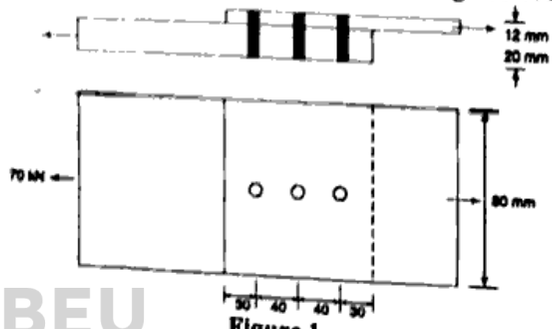


Figure 1

(b) Define effective throat thickness. Write down the types of weld joint. [7]

Q.4 (a) Determine the design bending strength of ISLB 350 @ 406 N/m considering the beam to be laterally supported. The design shear force is less than the design shear strength. [7]

(b) Explain web buckling and web crippling along with a neat sketch. [7]

Q.5 (a) A single unequal angle 100 x 75 x 8 is connected to a 12-mm thick gusset plate at the ends with six 18 min-diameter bolts to transfer tension as shown in following figure 2. Determine the design tensile strength of the angle if the yield and the ultimate stress of steel used are 250 MPa and 410 MPa: (i) if the gusset is connected to the 100-mm leg. [14]

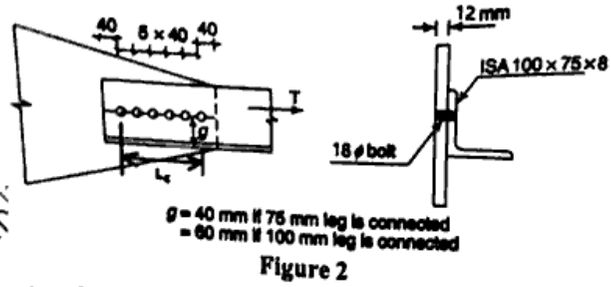


Figure 2

Q.6 Design a batten system for a 12-m long column to carry a factored axial load 1200kN. The column is restrained in position but not in direction at both ends. Assume that the two channels are kept back-back. [14]

Q.7 (a) Discuss in detail (i) Stiffened and unstiffened seat connections (ii) Type 1 and type 2 bracket connections. [7]

(b) How do fillet welds differ from butt welds, and in what circumstances is a fillet weld typically preferred? [7]

Q.8 A simply supported beam of span 6 m supports a reinforced concrete slab. The compression flange of the beam is restrained due to its connection with the slab. The beam is subjected to a dead load of 25 kN/m and an imposed load of 20 kN/m. Design the beam. [14]

Q.9 Design the base plate for the column of length 4m subjected to a factored moment of 45 kNm and a factored axial load of 500 kN. The column size is ISHB 250. The cube compressive strength of concrete in the foundation is fck = 25 N/mm^2. Use grade 410 steel. [14]

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